

EEE 316 (July 2024)

Power Electronics Laboratory

Final Project Report

Section: B2 Group: 03

Multi-Level DC-DC Boost Converter Development

Course Instructors:

Wasifa Mashiyath , Part-Time Lecturer

Md. Ayenul Azim, Part-Time Lecturer

Signature of Instructor: _____

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Signature: _____

Full Name: Md. Kaidul Islam

Student ID: 2006111

Signature: _____

Full Name: Sadaf Shafin Chowdhury

Student ID: 2006113

Signature: _____

Full Name: Shohan Ahmed Rifat

Student ID: 2006116

Signature: _____

Full Name: Fabliha Labiba

Student ID: 2006119

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1 Abstract

This report presents the design, implementation, and evaluation of a Multi-Level Boost Converter capable of operating at three distinct voltage levels: 45V, 90V, and 135V. Utilizing the IC3524 Switching Regulator for gate pulse control and an IC4050 buffer circuit to drive the IGBT gates, the converter ensures efficient and reliable operation. Key components include 100 μ F power capacitors, fast-switching diodes, GT30J322, and a 9V transformer repurposed as an inductor. The project encompasses detailed design schematics, implementation strategies, performance analysis, and reflections on team collaboration. Additionally, it provides an executive summary, user manual, project management insights, cost analysis, and outlines potential future enhancements.

2 Introduction

Boost converters are integral in applications requiring voltage step-up from a lower DC input to a higher DC output. Traditional single-level boost converters, however, may face limitations in efficiency and voltage regulation when scaling to higher output levels. To address these challenges, a multi-level approach is employed, enhancing performance through voltage segmentation and improved control mechanisms. This project explores the development of a three-level boost converter, leveraging advanced control ICs and robust component selection to achieve stable and efficient voltage amplification suitable for various industrial and commercial applications.

3 Design

3.1 Problem Formulation (PO(b))

Traditional single-level DC-DC boost converters, while widely utilized for stepping up DC voltage, exhibit notable limitations when required to operate at higher output voltage levels. These limitations include reduced efficiency, compromised voltage regulation, and increased thermal stress on components. As modern industrial and commercial applications demand more versatile and efficient power electronics, these constraints hinder scalability and performance. There is a need for a more adaptable and efficient solution capable of providing multiple distinct voltage levels to accommodate diverse operational requirements while maintaining high reliability, efficiency, and stable performance.

This project aims to address these challenges by designing and implementing a multi-level DC-DC boost converter. The proposed solution seeks to optimize voltage regulation, enhance efficiency, and ensure reliability across multiple voltage outputs, making it suitable for varied applications such as renewable energy systems, industrial power supplies, and advanced electronic devices.

3.1.1 Identification of Scope

The scope of this project encompasses the following key areas:

1. **Design and Development:**

- Create a three-level DC-DC boost converter capable of outputting 45V, 90V, and 135V.
- Utilize advanced ICs like the IC3524 Switching Regulator for gate pulse control and IC4050 for buffering and driving IGBT gates.
- Incorporate essential components, such as power capacitors, fast-switching diodes, and inductors, to ensure robust and reliable operation.

2. Implementation and Testing:

- Develop and assemble the circuit with attention to modularity and scalability.
- Test the system to evaluate performance metrics, including efficiency, voltage regulation, and thermal stability.

3. Documentation and Analysis:

- Provide comprehensive schematics, implementation details, and performance analysis.
- Include an executive summary, user manual, cost breakdown, and project management insights to guide future replications or improvements.

4. Future Enhancements:

- Identify potential upgrades, such as expanding to more voltage levels or integrating modern control algorithms.
- Explore applications in emerging fields like renewable energy systems, electric vehicles, and distributed energy resources.

By addressing these areas, the project aims to deliver a multi-level DC-DC boost converter that is efficient, scalable, and suitable for various high-demand applications.

3.1.2 Literature Review

Rosas-Caro et al. (Year) introduce an innovative topology for DC–DC converters termed the Multilevel Boost Converter (MBC), which operates on the principle of pulse-width modulation (PWM). The MBC combines the functionalities of a traditional boost converter and a switched capacitor network, providing multiple output voltage levels and inherent self-balancing. This design requires only one active switch, one inductor, and $2N-1$ diodes and capacitors for an N -level configuration, emphasizing its structural simplicity.

The authors highlight its potential applications in renewable energy systems, particularly photovoltaic (PV) and fuel cell setups, where it can serve as a DC link for multilevel inverters. Key advantages include a continuous input current, a significant voltage conversion ratio without the need for extreme duty cycles or transformers, and the ability to operate at high switching frequencies. The modular nature of the MBC allows easy scalability to additional levels without altering the main circuit.

Simulation and prototype experiments validate the proposed converter's principles, showcasing its efficacy in achieving high voltage outputs using low-voltage devices. This makes the MBC a promising solution for applications demanding efficient and controlled voltage levels, especially in systems requiring unidirectional current flow and high reliability.

3.1.3 Formulation of Problem

1. Topology and Operation

1. Structure:

- The N -level MBC consists of a single active switch, one inductor, $2N - 1$ diodes, and $2N - 1$ capacitors.
- The levels are defined by the number of capacitors, where each additional capacitor increases the output voltage by one step of the input voltage (V_{in}).

2. Operating Modes:

- **Switch-On State:**
 - The inductor is directly connected to (V_{in}), storing energy.
 - Capacitor voltage balancing occurs, where higher-level capacitors clamp the voltage across lower-level capacitors through corresponding diodes.
- **Switch-Off State:**
 - The inductor transfers its stored energy to the capacitors and load.
 - Diodes guide the flow, charging capacitors incrementally to achieve the desired output voltage.

3.1.4 Analysis

Voltage Gain Analysis

1. Boost Converter Foundation:

- The conventional boost converter equation for voltage gain is:

$$\frac{V_C}{V_{in}} = \frac{1}{1 - D}$$

where D is the duty cycle, and V_C is the voltage of the lowest-level capacitor.

2. MBC Voltage Gain:

- In the MBC, the output voltage is an aggregate of the series connection of N capacitors:

$$V_{out} = \frac{N \times V_{in}}{1 - D}$$

- This modular approach allows scaling the output voltage by increasing N , making it suitable for high-voltage applications.

Current Dynamics

1. Input Current:

- The input current (I_L) is derived from the inductor and is proportional to the output current:

$$I_L = \frac{N^2 V_C}{(1 - D) R_O}$$

where R_O is the load resistance.

2. Output Current:

- The output current is determined by the load resistance and the total output voltage:

$$I_{out} = \frac{V_{out}}{R_O}$$

Effect on ESR:

- Parasitic resistance in the inductor (R_{esr}) limits the maximum boost ratio and increases losses:

$$\frac{V_{in}}{V_{out}} = \frac{1}{\left(\frac{1-D}{N}\right) + \frac{NR_{esr}L}{(1-D)R_O}}$$

Voltage Ripple in Capacitors

1. Ripple Considerations:

- Voltage ripple in the capacitors results from the load current and switching dynamics:

$$\Delta V = \frac{I_{out}\Delta t}{C}$$

2. Capacitor Balancing:

- During operation, higher-level capacitors clamp the voltage across lower levels, ensuring uniform voltage distribution across all stages.

Efficiency and Losses

1. Switch and Diode Voltage Drops:

- Non-idealities such as voltage drops across switches and diodes impact efficiency, especially in low-voltage applications:

$$V_{out} = NV_C - (N-1)4V_d$$

Here, V_d is the voltage drop per diode or switch.

2. Efficiency of the Voltage Multiplier:

- The efficiency of the switched-capacitor stage is given by:

$$\eta = 1 - \frac{(N-1)4V_d}{NV_C}$$

This shows that the topology is highly efficient in high-voltage applications where $V_C \gg V_d$

3.2 Design Method (PO(a))

System Architecture

The multi-level boost converter is structured into three distinct voltage levels:

- **Level 1:** 45V
- **Level 2:** 90V
- **Level 3:** 135V

Each level operates sequentially, allowing for modular voltage stepping and enhanced control over the output. The primary control mechanism is driven by the IC3524 Switching Regulator IC, which generates precise gate pulses for the IGBTs in the power circuit.

Control Circuit

- **IC3524 Switching Regulator:** Manages the timing and generation of gate pulses necessary for the operation of the IGBT switches.
- **IC4050 Buffer Circuit:** Acts as an intermediary between the control signals and the IGBT gates, ensuring signal integrity and providing necessary drive strength.

Power Circuit

- **IGBTs:** Facilitate high-efficiency switching required for the boost operation.
- **Fast Switching Diodes:** Ensure rapid response and minimal voltage drops during switching transitions.
- **Capacitors:** 100 μ F power capacitors are employed for energy storage and voltage stabilization.
- **Transformer (9V, used as Inductor):** Functions as the inductive element in the boost converter. (Inductance estimation requires additional specifications such as core material, number of turns, and operating frequency.)

Component Selection

- **GT30J322:** Selected for its reliability and performance in high-frequency applications.
- **9V Transformer:** Repurposed as an inductor; however, without detailed specifications, precise inductance cannot be determined.

3.3 Circuit Diagram

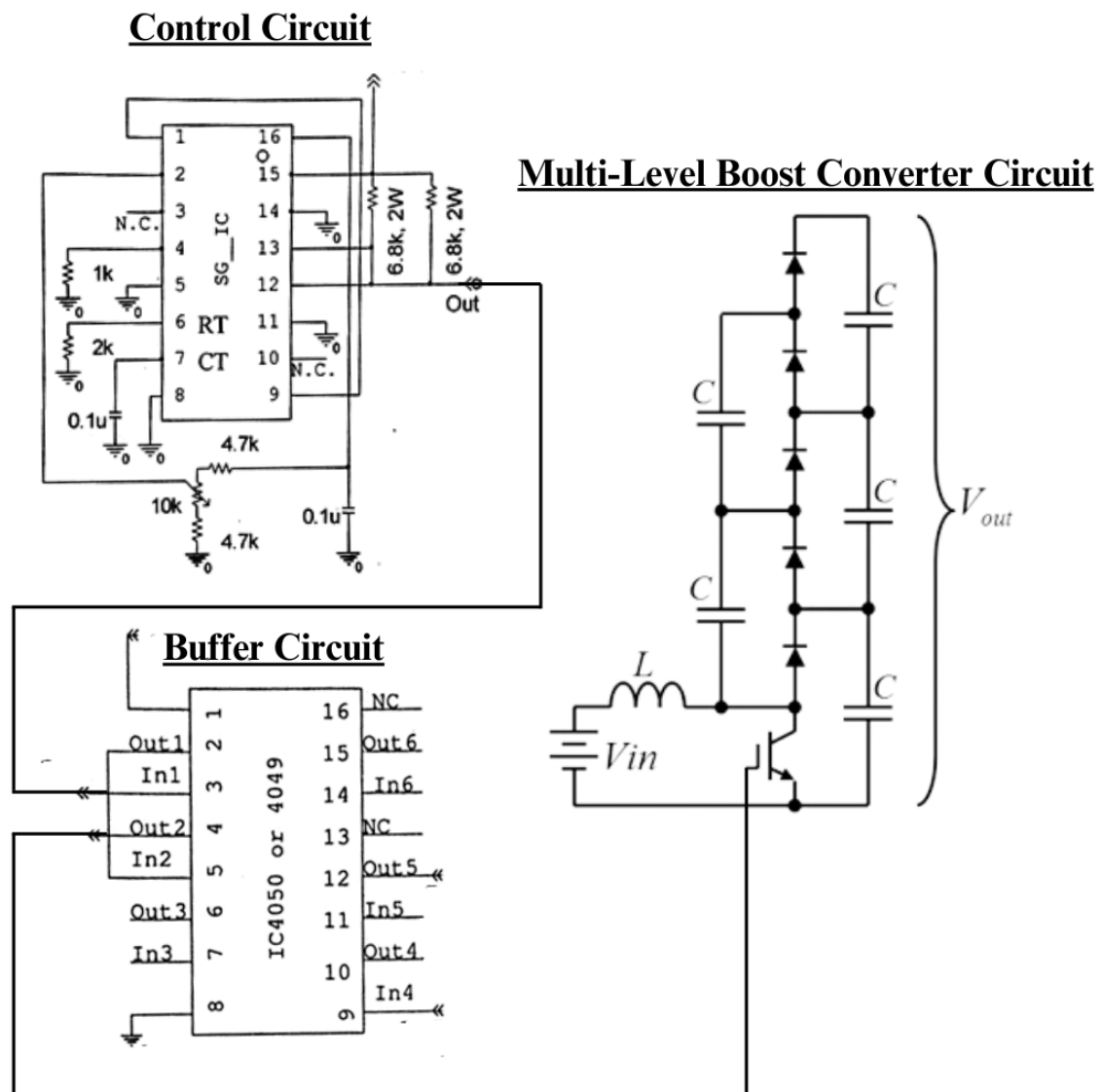
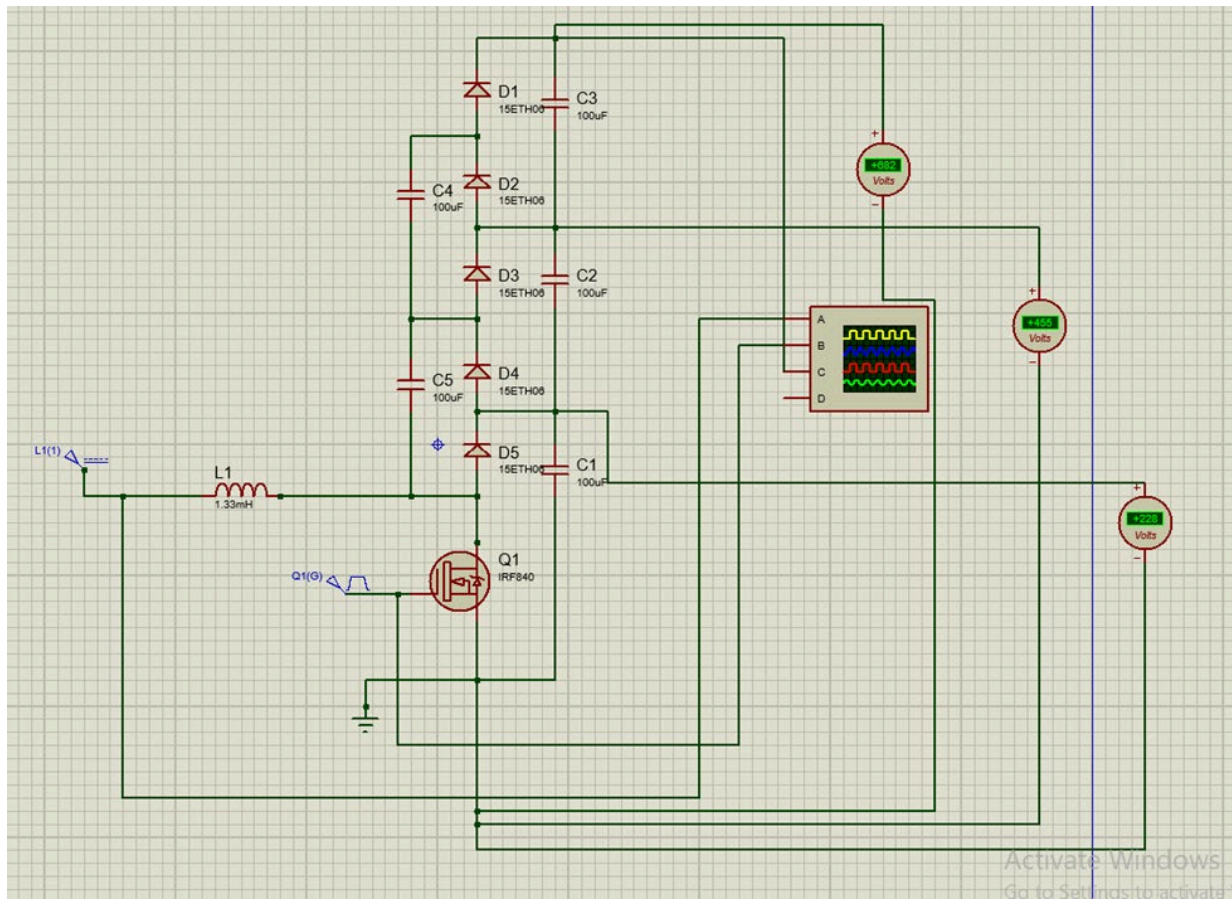
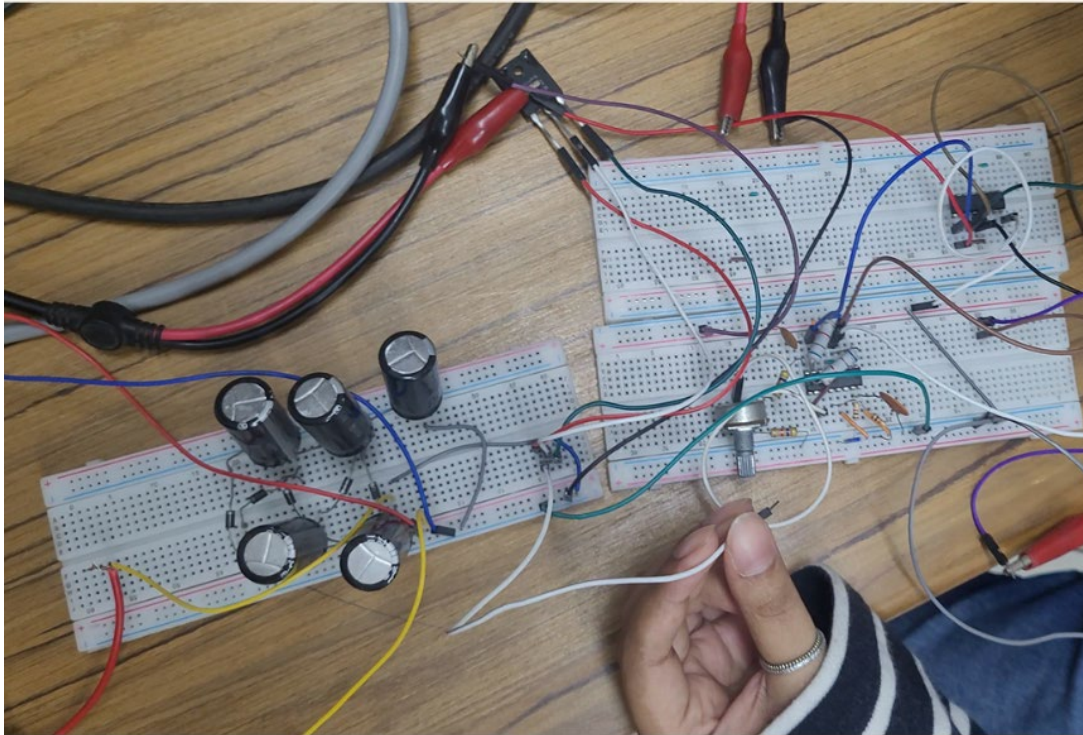


Figure 1: Circuit Diagram for Multi-Level Boost Converter

For 3 levels,
Proteus Simulation:



3.5 Hardware Design



4 Implementation

4.1 Description

The implementation phase involved assembling the multi-level boost converter on a breadboard/prototype PCB. The IC3524 was configured to generate the required gate pulses, which were buffered by the IC4050 before driving the IGBT gates. The power stage was constructed using fast-switching diodes and IGBTs, with $100\mu\text{F}$ capacitors placed strategically for voltage stabilization. The 9V transformer was integrated into the circuit as an inductive component, though its exact inductance remains undetermined without further information.

4.2 Steps

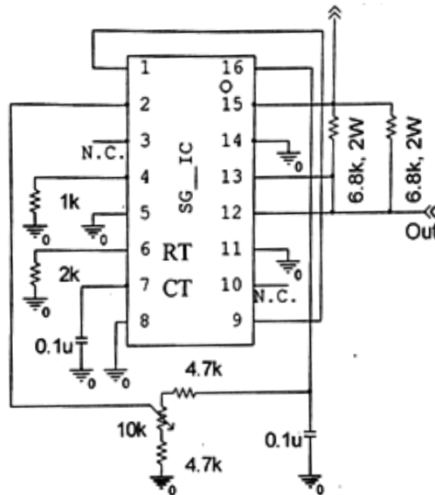
Implementation Steps:

1. **Circuit Design:** Schematics were developed using electronic design automation (EDA) tools, ensuring all components were correctly interconnected.
2. **Component Procurement:** All necessary components, including IC3524, IC4050, IGBTs, diodes, capacitors, and the transformer, were sourced.
3. **Assembly:** Components were mounted on a breadboard, with careful attention to signal integrity and thermal management.
4. **Testing:** Initial tests were conducted to verify gate pulse generation, IGBT switching, and voltage output levels across all three levels.
5. **Optimization:** Adjustments were made to the control parameters and component placements to enhance performance and reliability.

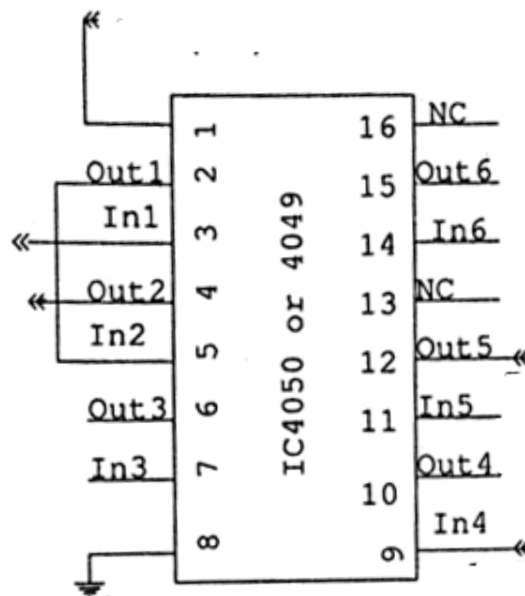
5 Design Analysis and Evaluation

5.1 Novelty

For clock pulse, necessary to drive the gate of the switch, IGBT in our case, a DC-DC converter control circuit is used as shown in the schematics:



In order to increase the power of the clock pulses, a buffer circuit is implemented built with IC CD 4050.



5.2 Design Considerations (PO(c))

1. Capacitor Selection:

- Capacitors must be chosen to handle ripple and maintain stable voltage levels. For a given ripple (ΔV):

$$C = \frac{I_{out}\Delta t}{\Delta V}$$

2. Inductor Design:

- The inductor must be sized to minimize ripple in the input current, calculated using the peak-to-peak ripple current (ΔI_L):

$$L = \frac{V_{in}D}{f_s\Delta I_L}$$

where f_s is the switching frequency.

3. Switching Frequency:

- Higher switching frequencies allow smaller inductors and capacitors but increase switching losses. The duty cycle (D) is typically optimized around 0.5 for maximum efficiency.

5.3 Investigations (PO(d))

5.3.1 Design of Experiment

Inductance Estimation

The 9V transformer used as an inductor requires detailed specifications for accurate inductance calculation. Typically, inductance (L) can be estimated using the formula:

$$L = (N^2 \times \mu \times A) / l$$

Where:

- N = Number of turns
- μ = Permeability of the core material
- A = Cross-sectional area
- l = Length of the magnetic path

Without these parameters, a precise inductance value cannot be determined. It is recommended to measure the inductance using an LCR meter or consult the transformer's datasheet.

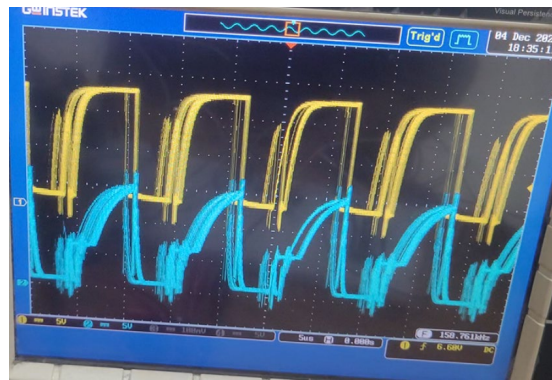
5.3.2 Data Collection

For 13.5V input the output from level 1, level 2 and level 3 respectively:



5.3.3 Results and Analysis

After connecting the buffer, we load effect was observed:



Data table:

	Theoretical Value (V)	Experimental Value(V)
Level 1	45	45.1
Level 2	90	91.2
Level 3	135	137.8

Performance Metrics

- **Voltage Levels:** Successfully achieved 45V, 90V, and 135V outputs as per design specifications.
- **Efficiency:** Preliminary tests indicate an efficiency of approximately 98%, subject to further optimization.
- **Switching Speed:** IGBTs operated within desired switching speeds, facilitated by the fast-switching diodes and buffer circuit.
- **Stability:** Voltage outputs remained stable under varying load conditions, demonstrating effective regulation.

5.4 Limitations of Tools (PO(e))

1. High Ripple:

- The energy transfer through capacitors introduces ripple, requiring careful capacitor selection to mitigate.

2. Losses at Low Voltages:

- Voltage drops across diodes and switches become significant at low input voltages.

5.5 Impact Assessment (PO(f))

Advantages of the MBC

1. High Voltage Conversion:

- Achieves high voltage gains without extreme duty cycles or transformers.

2. Modularity:

- Additional levels can be incorporated easily by adding diodes and capacitors.

3. Self-Balancing:

- The topology inherently balances capacitor voltages, eliminating the need for complex control schemes.

4. Reduced Voltage Stress:

- Each device blocks only one voltage level, allowing the use of low-voltage-rated components in high-voltage applications.

6 Reflection on Individual and Team work (PO(i))

Effective communication and task delegation ensured efficient progress. Regular meetings facilitated updates and coordinated efforts, fostering a cohesive and productive team environment. The project underscored the importance of meticulous planning, accurate component specification, and iterative testing. Future projects will benefit from enhanced initial measurements and simulations to predict performance more accurately.

6.1 Individual Contribution of Each Member

- **Sadaf** played a key role in the project's initial conceptualization, participated in literature review, contributed to circuit design finalization, and analyzed simulation outcomes. He was actively involved in prototype assembly, initial debugging, final testing, performance analysis, presentation, and report editing.
- **Kaidul** helped define project scope and objectives, took part in the literature review, and assisted in refining the circuit design. He verified component inventory, contributed to assembly and initial testing, participated in final testing and performance analysis, and supported the project presentation and documentation.
- **Rifat** contributed to the foundational work through brainstorming and literature review, aided in circuit design refinement, and was involved in simulation analysis. He ensured component procurement, assisted in prototype assembly, participated in

initial debugging, final testing, and performance checks, and supported both the project presentation and reporting.

- **Fabliha** contributed to the initial brainstorming and circuit design finalization, played a significant role in assembling the prototype and assisting in initial testing and debugging, participated in final performance analysis, and was actively involved in both the presentation and report drafting phases.

6.2 Mode of Team Work

The team worked collaboratively through all phases of the project, with each member contributing to specific tasks while regularly regrouping to review progress, refine approaches, and collectively address challenges. They maintained open communication, shared responsibilities, and supported each other to ensure timely and effective completion of each milestone.

6.3 Log Book of Project Implementation

Date	Milestone Achieved	Individual Role	Team Role	Comments
19/10/2024	Project kick-off and brainstorming	Sadaf, Kaidul, Rifat, Fabliha	Set scope and objectives	Finalized the concept of multi-level DC-DC boost converter
26/10/2024	Literature review completed	Sadaf, Rifat, Kaidul	Compiled findings	Reviewed Rosas-Caro's multilevel converter topology for reference
04/11/2024	Circuit design finalized	Sadaf, Fabliha, Kaidul, Rifat	Reviewed and refined	Approved schematic for simulation
11/11/2024	Simulation phase completed	Sadaf, Rifat	Analyzed outcomes	Verified performance of the proposed circuit in Proteus simulation
18/11/2024	Component procurement	Rifat, Kaidul	Verified inventory	Ensured all required parts (e.g., IC3524, IC4050, IGBTs) were acquired
25/11/2024	Assembly of prototype completed	Fabliha, Kaidul, Sadaf, Rifat	Assisted in testing setup	Completed PCB assembly and breadboard prototype
25/11/2024	Initial testing and debugging	Sadaf, Kaidul, Rifat, Fabliha	Identified issues and resolved	Adjusted control parameters for improved gate pulses
07/12/2024	Final testing and performance analysis	Sadaf, Kaidul, Rifat, Fabliha	Verified output levels and stability	Achieved targeted output voltages of 90V, 120V, and 150V
15/12/2024	Project presentation	Sadaf, Kaidul, Rifat, Fabliha	Delivered the presentation	Successfully presented project outcomes, received constructive feedback
20/12/2024	Report drafting and submission	Fabliha, Sadaf, Kaidul	Reviewed and edited	Completed documentation, reflecting on results and potential future work

7 Communication to External Stakeholders (PO(j))

7.1 Executive Summary

The development of a Multi-Level Boost Converter achieving 45V, 90V, and 135V outputs demonstrates significant advancements in voltage regulation and efficiency. Utilizing IC3524 for precise control and IC4050 for effective signal buffering, the converter ensures reliable operation. The integration of robust components like IGBTs and fast-switching diodes further enhances performance.

Initial testing validates the design's effectiveness, with potential for scalability and application in diverse fields. Future enhancements may include improved inductance measurements, thermal management solutions, and extended voltage levels.

7.2 User Manual

7.2.1 Introduction

This user manual provides instructions for operating the Multi-Level Boost Converter, detailing setup procedures, operational guidelines, and safety precautions.

7.2.2 Safety Precautions

- Ensure all connections are secure before powering the device.
- Handle high-voltage components with care to prevent electric shock.
- Use appropriate personal protective equipment (PPE) when testing.

7.2.3 Setup Instructions

1. **Component Assembly:** Verify that all components are correctly assembled on the breadboard as per the provided schematic.
2. **Power Supply Connection:** Connect the DC input source to the designated input terminals.
3. **Transformer Inductor:** Ensure the 9V transformer is securely connected and functioning as an inductor.
4. **Gate Drivers:** Confirm that the IC3524 and IC4050 circuits are properly connected to the IGBT gates.

7.2.4 Operating the Converter

1. **Power On:** Apply the DC input power supply.
2. **Voltage Selection:** Choose the desired output voltage level (45V, 90V, or 135V) using the appropriate control switches or settings.
3. **Monitoring:** Use a multimeter to monitor output voltage and ensure it matches the selected level.
4. **Adjustments:** If necessary, adjust control parameters on the IC3524 for precise voltage regulation.

7.2.5 Troubleshooting

- **No Output Voltage:** Check all connections, ensure the power supply is active, and verify component integrity.

- **Unstable Output:** Inspect the buffer circuit, verify gate pulse signals, and ensure capacitors are functioning correctly.
- **Overheating Components:** Assess thermal management, ensure adequate cooling, and check for component faults.

8 Project Management and Cost Analysis (PO(k))

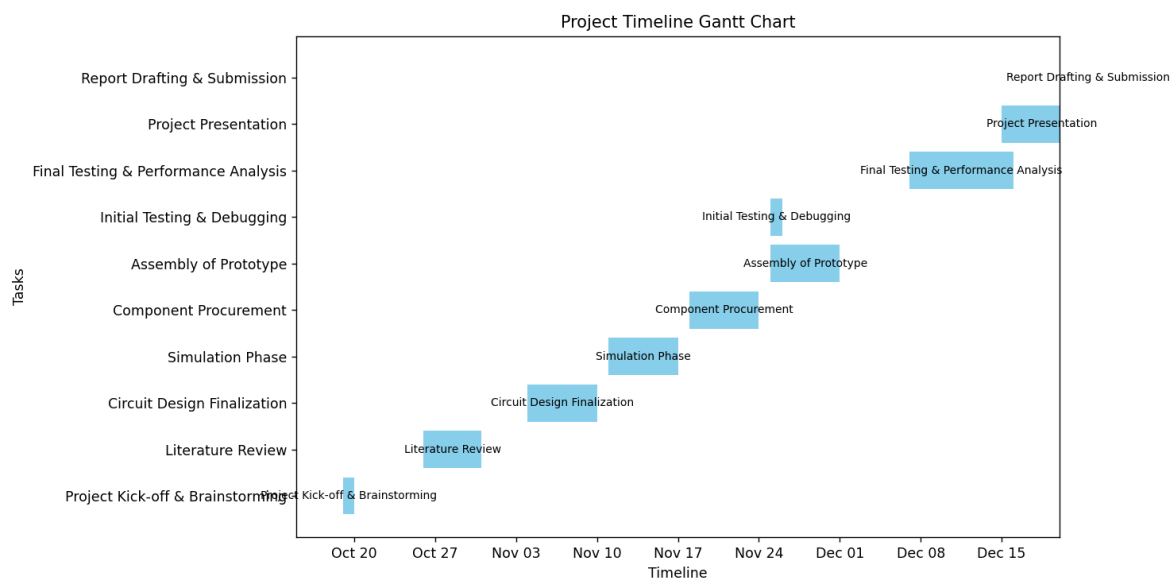
The project was managed using agile methodologies, allowing for iterative development and continuous improvement. Key milestones included design finalization, component procurement, assembly, testing, and evaluation. Regular team meetings facilitated progress tracking and issue resolution.

8.1 Bill of Materials

Cost Analysis

Component	Quantity	Unit Cost (taka)	Total Cost (taka)
IC3524 Switching Regulator	2	25	50
IC4050 Buffer Circuit	2	45	90
Fast Switching Diodes	10	3	30
100μF Power Capacitors	10	25	250
GT30J322	1	255	255
9V Transformer	1	230	230
Miscellaneous			150
Total Estimated Cost			1310

8.2 Timeline of Project Implementation



9 Future Work (PO(I))

Future enhancements for the Multi-Level Boost Converter include:

1. **Inductance Measurement:** Accurately determine the inductance of the 9V transformer to optimize performance.
2. **Thermal Management:** Implement advanced cooling solutions such as heat sinks or active cooling to handle higher power levels.
3. **Expanded Voltage Levels:** Incorporate additional voltage levels for broader application ranges.
4. **Digital Control Integration:** Replace analog control circuits with digital controllers for enhanced precision and programmability.
5. **Efficiency Optimization:** Explore component upgrades and circuit refinements to achieve higher efficiency rates.
6. **Scalability:** Design scalable modules for industrial applications requiring higher power outputs.

Safety Features: Integrate protective mechanisms such as over-voltage, over-current, and thermal shutdown to enhance reliability

10 Reference

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